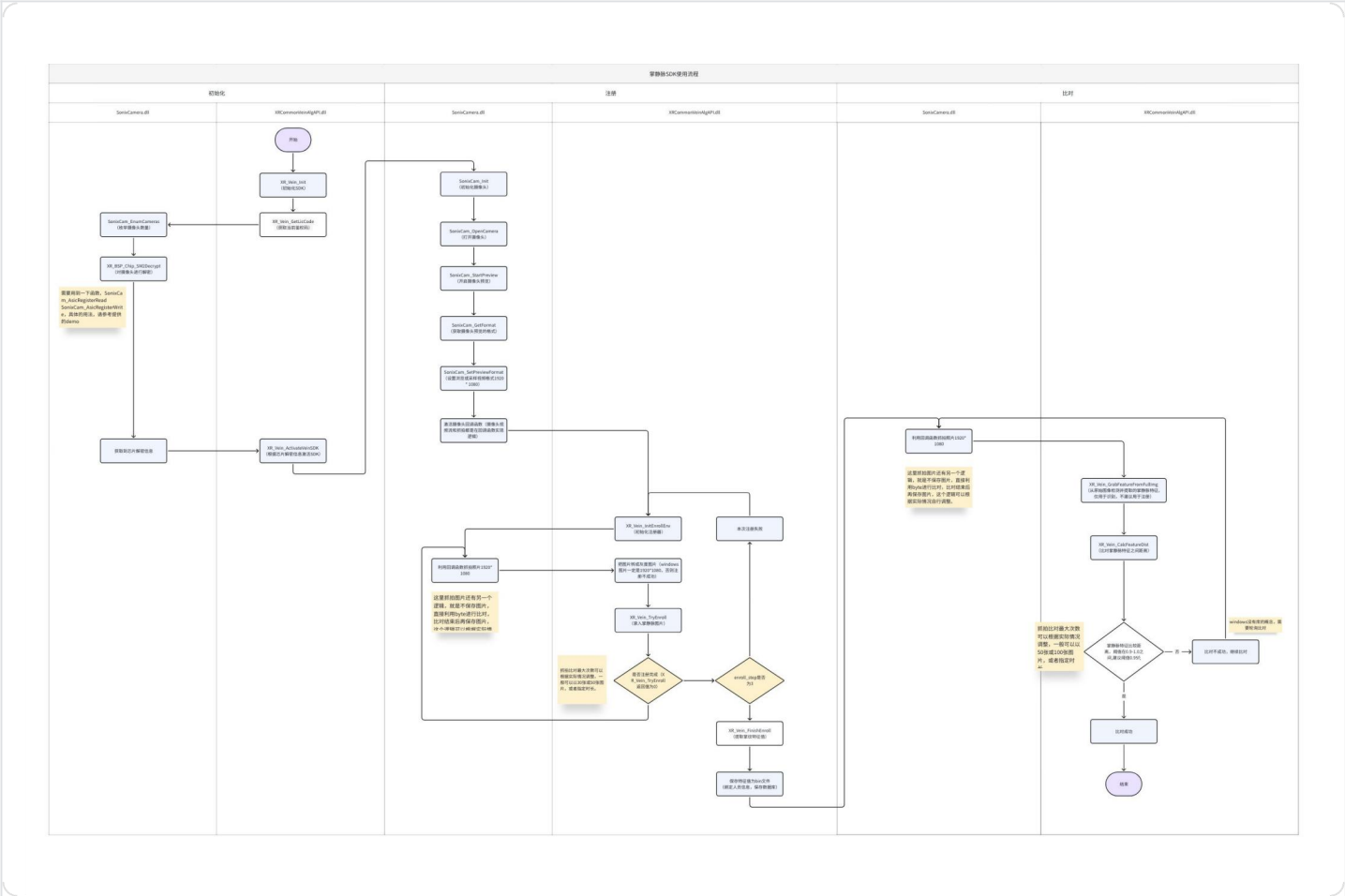
Palm Vein Module SDK User Documentation (Java)

1. Instructions

Palm vein module is divided into 2 parts , one is camera related (SonixCamera.dll), the other is algorithm related (XRCommonVeinAlgAPI.dll), in essence, it is necessary to reference these two dlls for operation.

Recommended usage process:



2. dll description

2.1 SonixCamera.dll

Please see the attachment for details.

2.2 XRCommonVeinAlgAPI.dll

Please see the attachment for details.

3. Calling example

3.1 Put the file-related files into the specified folder

Can be adjusted according to actual conditions, here we take the program root directory as an example

3.2 Add to jna support

Code Block

```
< dependencies >
  < dependency >
    < groupId >net.java.dev.jna</ groupId >
    <artifactId> jna </artifactId>
    < version >5.12.1</ version >
  </ dependency >
  < dependency >
    < groupId >net.java.dev.jna</ groupId >
    <artifactId> jna - platform </artifactId>
    < version > 5.12.1 </ version >
  </ dependency >
</ dependencies >
```

3.3 References dll

Specific usage, please refer to the example

Code block

```
package org.example;

import com.sun.jna.*;
import com.sun.jna.platform.win32.WinDef;
import com.sun.jna.ptr.ByteByReference;
import com.sun.jna.ptr.IntByReference;
import com.sun.jna.ptr.LongByReference;
import com.sun.jna.ptr.PointerByReference;
import com.sun.jna.win32.StdCallLibrary;

import java.util.Arrays;
import java.util.List;
import java.util.concurrent.Executors;
import java.util.concurrent.ScheduledExecutorService;

/**
 * JNA interface of Sonix Lib camera library
```

```
 */
public interface Son ixCamera extends Library {
      SonixCamera INSTANCE = Native. load ("SonixCamera", SonixCamera.class);
}
```

Code Block

```
package org.example;

import com.sun.jna.Library;
import com.sun.jna.Native;
import com.sun.jna.Pointer;
import com.sun.jna.Structure ;
import com.sun.jna.ptr.IntByReference;
import com.sun.jna.ptr.PointerByReference ;

public interface XRCommonVei nAlgAPI extends Library {

// Loading dll files
      XRCommonVeinAlgAPI INSTANCE = Native.load("XRCommonVeinAlgAPI",
XRCommonVeinAlgAPI.class);

  }
```

3.4 Specific call

3.4.1 Initialize the SDK XR _ Vein _ Init

Code Block

```
// Initialize the camera
isOk = SonixCamera. INSTANCE . SonixCam_Init ();
```

3.4.2 Get the current authentication code XR _ Vein _ GetLicCode

Code Block

```
// Get the authorization code
ret = XRCommonVeinAlgAPI.INSTANCE.XR_Vein_GetLicCode(algHandle, licData,
licLen);
```

3.4.3 Enumerate the number of cameras SonixCam _ EnumCameras

Code Block

```
// Enumerate the number of device
boolean res = SonixCamera.INSTANCE.SonixCam_EnumCameras(devCount, camerasList,
1);
```

3.4.4 Decrypt the camera XR_BSP_Chip_SM 2 Decrypt

Code Block

```
// XR_BSP_Chip_SM2Decrypt
int decryptRet = XR_BSP_Chip_SM2Decrypt(licData, licLen.getValue(), dstData1,
outLen);
```

3.4.5 Activate SDK based on chip decryption information XR_Vein_ActivateVeinSDK

Code Block

```
// SDK Activation
ret = XRCommonVeinAlgAPI. INSTANCE . XR_Vein_ActivateVeinSDK (
algHand le, activationCode. getBytes (), activationCode. length ());
```

3.4.6 Initialize the camera SonixCam_Init ()

Code Block

```
SonixCamera.INSTANCE.SonixCam_Init();
```

3.4.7 Open the camera SonixCam _ OpenCamera

Code Block

```
api. SonixCam_OpenCamera (myCamerasList[ 0 ]. getPointer (), hWnd,
callback, Pointer. NULL )
```

3.4.8 Start camera preview SonixCam _ StartPreview

Code Block

```
boolean startPreview =
SonixCamera.INSTANCE.SonixCam_StartPreview(myCamerasList[0].getPointer());
```

3.4.9 Get the camera preview format SonixCam_GetFormat

Code Block

```
boolean startPreview =
SonixCamera.INSTANCE.SonixCam_StartPreview(myCamerasList[0].getPointer());
```

3.4.10 Set preview or sample video format 1920 * 1080 SonixCam_SetPreviewFormat

Code Block

```
// Set the preview format to 1920 * 1 080
SonixCamera. INSTANCE . SonixCam_SetPreviewFormat (myCamerasList[ 0 ].
getPointer (), 1 , new WinDef. HWND (Pointer. NULL ) );
```

3.4.11 Initialize the registrar XR _ Vein _ InitEnrollEnv

Code Block

```
XRCommonVeinAlgAPI. INSTANCE . XR_Vein_InitEnrollEnv (alg Handle);
```

3.4.12 Enroll palm vein image

XR _ Vein _ TryEnroll

Code Block

```
int ret = XRCommonVeinAlgAPI.INSTANCE.XR_Vein_TryEnroll(
        algHandle, grayImg, img.getHeight(), img.getWidth(),
        1, enrollStep, capTip, palmInfo, highBright);
```

3.4.13 Extract palmprint feature values

XR _ Vein _ FinishEnroll

Code Block

```
int ret = XRCommonVeinAlgAPI. INSTANCE . XR_Vein_FinishEnroll
    (algHandle, roiImgBuf, roiLen, featBuf, featLenRef);
```

3.4.14 Palm vein features detected and extracted from the original image XR_Vein_GrabFeatureFromFullImg

Code Block

```
XR_Vein_GrabFeatureFromFullImg
```

3.4.15 Compare the distances between palm vein features XR_Vein_CalcFeatureDist

Code Block

```
XR_Vein_CalcFeatureDist
```